

MARS: Week two meeting slides

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FEATURE TABLE INTERPRETATION

Understanding what each row and column represents, and what we aim for in feature design.



WHAT THE TABLE MEANS



Each row represents one video frame.

The table is organised in time order (e.g., $t = 0, 1, 2, \dots$).



Each column represents a handcrafted feature extracted from that frame.

Each column has a specific, measurable meaning.



Together, the feature values form a feature vector describing the frame.

This vector summarises the visual content of the frame.



The *label* indicates whether the frame is **normal or **anomalous**.**

This is the ground truth used for training or evaluation.



DESIRABLE PROPERTIES OF EXTRACTED FEATURES



1. Expressive

Capture sufficient information to enable accurate downstream tasks such as classification and anomaly detection.



2. Explainable

Each feature has a clear, interpretable meaning (e.g., average pixel intensity, frame difference, or bright pixel count), allowing model decisions to be understood and validated.

EXAMPLE: FEATURE TABLE OVER TIME

Hand-crafted features extracted from each frame of a video sequence

frame (t)	roi_mean_intensity (μ_R)	bright_pixel_count (B_t)	frame_difference (D_t)	edge_density (E_t)	motion_ratio (M_t)	entropy (H_t)	label
t = 0	98	412	0.00	0.21	0.01	4.83	normal
t = 1	99	418	0.04	0.22	0.02	4.81	normal
t = 2	180	890	0.87	0.53	0.48	6.14	anomaly
t = 3	176	872	0.81	0.51	0.45	6.07	anomaly
t = 4	101	425	0.05	0.23	0.03	4.86	normal

FEATURE DESCRIPTIONS

- μ_R Mean intensity inside the region of interest (ROI).
- B_t Number of bright pixels above a threshold τ .
- D_t Mean absolute frame difference w.r.t. previous frame.
- E_t Edge density: ratio of edge pixels (Canny) to total pixels.
- M_t Motion ratio: ratio of moving pixels to total pixels.
- H_t Shannon entropy of the grayscale image.
- label Ground truth: normal or anomaly.

INTERPRETATION

-  Frames $t = 0, 1, 4$ have low motion and stable brightness → normal.
-  Frames $t = 2, 3$ show large changes in multiple features → anomaly.
-  Multiple simple features together provide a richer, more robust signal than any single feature alone.
-  These features can be used for thresholding, rule-based alarms, or as input to ML models for anomaly detection.

Existing feature-extraction approaches for images and video

Math -> Typical use -> Explainability

1. ROI MEAN INTENSITY / COLOUR

The baseline: summarise all pixels inside a chosen region

MATHEMATICAL DEFINITION

Let I be an image and $R \subseteq \Omega$ a region of interest (ROI), where Ω is the set of all pixel locations.

For a grayscale image $I(p)$:

$$\mu_R = \frac{1}{|R|} \sum_{p \in R} I(p)$$

$I(p)$	intensity at pixel p
R	ROI (set of pixels)
$ R $	number of pixels in R
μ_R	mean intensity in R

For a colour image in RGB:

$$\mu_R = \begin{bmatrix} \mu_R^{(R)} \\ \mu_R^{(G)} \\ \mu_R^{(B)} \end{bmatrix} = \begin{bmatrix} \frac{1}{|R|} \sum_{p \in R} I_R(p) \\ \frac{1}{|R|} \sum_{p \in R} I_G(p) \\ \frac{1}{|R|} \sum_{p \in R} I_B(p) \end{bmatrix}$$

I_R, I_G, I_B are the red, green and blue channel intensities.

WHAT DOES IT CAPTURE?



- Average brightness (grayscale) inside the ROI.
- Average colour (RGB) describing the dominant colour of the region.
- Insensitive to spatial arrangement—only overall mean is kept.

TYPICAL USAGE



Quality control

- Check if a region is too bright/dark.
- Detect illumination drift over time.



Medical / microscopy

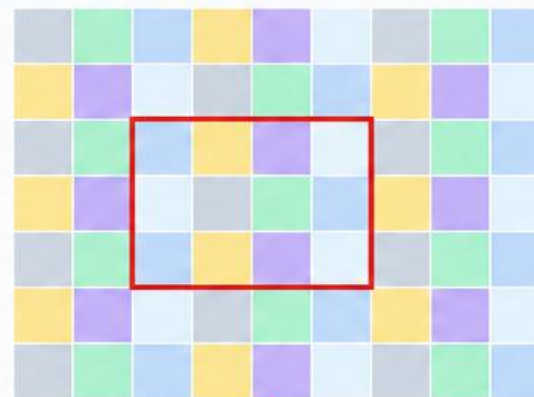
- Measure average intensity of cells / tissues.
- Track changes in staining or fluorescence.



Quick statistics

- Compute summary features for many ROIs.
- Fast baseline for downstream models.

HOW IT WORKS (VISUAL)



ROI (R)

Set of pixels inside the red rectangle

Output feature

Single summary vector

Grayscale: μ_R (scalar)

RGB: $\mu_R = [\mu_R^{(R)}, \mu_R^{(G)}, \mu_R^{(B)}]^T$

Dimension: 1 (gray) or 3 (RGB)

ADDITIONAL NOTES

- Very fast to compute: $O(|R|)$.
- Not invariant to changes in lighting (e.g., shadows, exposure).
- Loses all spatial information and texture/shape cues.
- Can be extended with other simple statistics:
 - Variance: $\sigma_R^2 = \frac{1}{|R|} \sum_{p \in R} (I(p) - \mu_R)^2$
 - Min / Max: $\min_{p \in R} I(p), \max_{p \in R} I(p)$
 - More channels: HSV, Lab, YCbCr, etc.

EXPLAINABILITY

High

Directly interpretable as the average pixel value(s) in a named region; easy to audit and reason about.



REFERENCE

OpenCV docs, `cv::mean`; Szeliski, R. (2022). Computer Vision: Algorithms and Applications (2nd ed.). Springer.

2. FRAME DIFFERENCE

The first video feature: change between neighbouring frames

MATHEMATICAL DEFINITION

Let $I_t(x, y)$ be the intensity of pixel at location (x, y) in frame t (grayscale image or use per-channel for colour).

1) Frame difference (absolute difference):

$$D_t(x, y) = |I_t(x, y) - I_{t-1}(x, y)|$$

2) Motion mask from threshold τ :

$$M_t(x, y) = \begin{cases} 1, & \text{if } D_t(x, y) > \tau \\ 0, & \text{otherwise} \end{cases}$$

Where:

$I_t(x, y)$: intensity at pixel (x, y) in frame t

$I_{t-1}(x, y)$: intensity at pixel (x, y) in previous frame $t - 1$

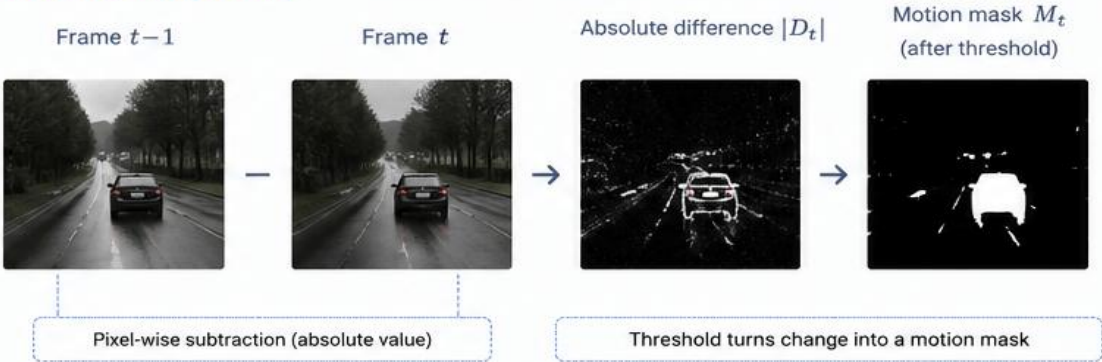
$D_t(x, y)$: absolute frame difference

$M_t(x, y)$: binary motion mask

τ : threshold controlling sensitivity

Note: For colour videos, compute per channel (R,G,B) or convert to grayscale first.

HOW IT WORKS (VISUAL)



WHAT DOES IT CAPTURE?



- Highlights pixels that changed between two consecutive frames.
- Sensitive to any intensity change (motion, lighting, noise).
- Produces a simple binary mask of moving regions.

TYPICAL USAGE

Motion detection

- Detect moving objects against a static camera.
- Simple surveillance motion alarms.

Event triggering

- Trigger recording or alerts before heavier models run.
- Reduce compute.

Foreground proposal

- Generate candidate regions for further processing (e.g., tracking, detection, recognition).

KEY LIMITATIONS



- Change $\neq$ object identity: cannot tell what changed, only that a change occurred.
- Sensitive to illumination changes (flicker, shadows, headlights).
- Fails with camera motion (global movement causes large differences).
- No memory: compares only two frames; may miss slow motions or intermittent changes.

PRACTICAL EXTENSIONS



- Apply Gaussian blur before differencing to reduce noise.
- Use adaptive or running background: $B_t = (1 - \alpha)B_{t-1} + \alpha I_t$ and compute $|I_t - B_t|$.
- Morphological operations (open/close) to clean the mask.
- Use per-pixel or local adaptive threshold instead of a global τ .

EXPLAINABILITY

High

Very interpretable: each bright pixel in D_t corresponds to a visible change. Easy to understand, visualize, and debug. Failure modes are intuitive (shadows, noise, camera shake, lighting).

AT A GLANCE (SUMMARY)

$$\begin{array}{ccccccc} I_{t-1}(x, y) & - & I_t(x, y) & | & | \cdot | & > \tau & \rightarrow & M_t(x, y) \in \{0, 1\} \\ \text{frame } t-1 & & \text{frame } t & & \text{absolute} & \text{threshold} & & \text{motion mask} \\ & & & & \text{difference} & & & \end{array}$$



REFERENCE

Zivkovic, Z. (2004). Improved adaptive Gaussian mixture model for background subtraction. In *Proceedings of the 17th International Conference on Pattern Recognition (ICPR)*.
Wren, C., Azarbayejani, A., Darrell, T., & Pentland, A. (1997). Pfnder: Real-time tracking of the human body. *IEEE Transactions on Pattern Analysis and Machine Intelligence*.

Szeliski, R. (2022). *Computer Vision: Algorithms and Applications* (2nd ed.). Chapters on motion estimation and video analysis.

3. BRIGHT PIXEL COUNT

Simple intensity-based feature: how many pixels are “bright” in a frame or region

MATHEMATICAL DEFINITION

Let $I_t(x, y)$ be the intensity of pixel (x, y) at time (frame) t .

Given a brightness threshold τ , define the indicator function:

$$\mathbf{1}[I_t(x, y) > \tau] = \begin{cases} 1, & \text{if } I_t(x, y) > \tau \\ 0, & \text{otherwise} \end{cases}$$

The bright pixel count for the whole frame is:

$$B_t = \sum_{x=1}^H \sum_{y=1}^W \mathbf{1}[I_t(x, y) > \tau]$$

For a region of interest (ROI) $R \subseteq \Omega$ (set of pixel coordinates):

$$B_t(R) = \sum_{(x, y) \in R} \mathbf{1}[I_t(x, y) > \tau]$$

Where:

$I_t(x, y)$ Intensity at pixel (x, y) in frame t (grayscale)

H, W Frame height and width

τ Brightness threshold (user-defined)

$\mathbf{1}[\cdot]$ Indicator function

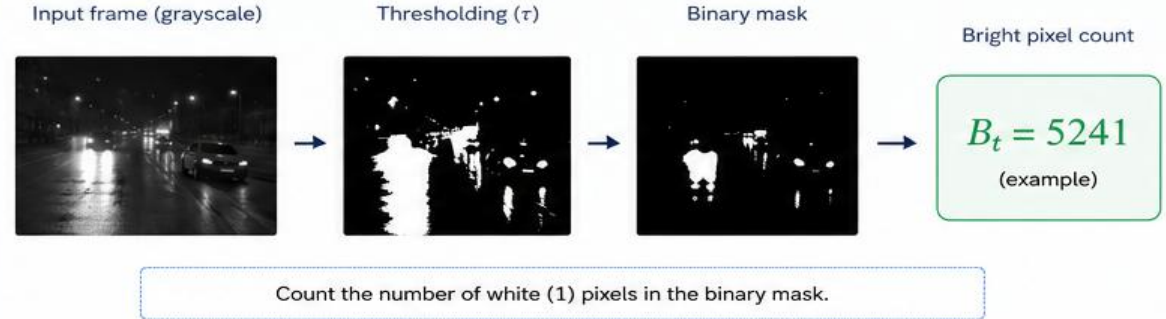
B_t Bright pixel count in the entire frame

$B_t(R)$ Bright pixel count in ROI R



For colour images, compute on a luminance channel (e.g., Y in YCbCr or grayscale conversion).

HOW IT WORKS (VISUAL)



WHAT DOES IT CAPTURE?



Overall brightness / amount of light in the scene.



Activity level (e.g., headlights at night, sparks, flames, reflections).



Useful as a simple scalar feature for monitoring or as an input to rule-based systems.



Very fast to compute and easy to interpret.

KEY LIMITATIONS



- Depends heavily on the threshold τ (too low → noisy, too high → misses details).
- Sensitive to global illumination changes (day/night, auto-exposure).
- Does not provide any spatial information (where the bright pixels are).
- Cannot distinguish between different causes of brightness.

TYPICAL USAGE



Surveillance

- Detect headlights, flashlights, or explosions.
- Trigger recording under low light conditions.



Lighting / Quality Control

- Monitor scene illumination levels.
- Detect abnormal brightening (e.g., sparks, glare).



Event Detection

- Simple event triggers (e.g., fire, lightning).
- Input feature for anomaly detection models.

EXPLAINABILITY

High

- ✓ Directly interpretable: the count of pixels whose intensity exceeds τ .
- ✓ Easy to audit and reason about.
- ✓ Simple and transparent; failure modes are intuitive.

PRACTICAL NOTES



- Choose τ based on the application (e.g., Otsu's method, percentile, or domain knowledge).
- Apply Gaussian blur or median filter before thresholding to reduce noise.
- Can be extended with weights: $B_t^{(w)} = \sum_{x,y} w(x, y) \mathbf{1}[I_t(x, y) > \tau]$ (e.g., emphasize certain regions).
- Works best in relatively static cameras or with background stabilisation.



REFERENCE

Haralick, R. M., Sternberg, S. R., & Zhuang, X. (1987). Image analysis using mathematical morphology. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 9(4), 532–550.

Szeliski, R. (2022). *Computer Vision: Algorithms and Applications* (2nd ed.). Springer. (Chapter on image filtering and thresholding).

PROJECT TIMELINE

A rough first-stage plan for the first few weeks of the project.



WEEK 1-2



Literature Review & Feature Comparison

- Conduct a literature review, leading to a features comparison table (the reviewed methods are expected to be summarized in a table).



WEEK 2-3



Implement ~10 Representative Features

- Implement ~10 representative, explainable features; show how each is extracted from frames into a time series (CSV).



WEEK 3-4



First Test on a Public Dataset

- Run a first test on a public video dataset (using its own labels) with a simple classifier, to see which features are actually useful.



BEYOND WEEK 4



Next Steps (Beyond the First Few Weeks)

- The goal of the first few weeks isn't just to review the literature, it's to get from video to explainable feature time-series and show they're useful for a basic task.



OVERALL GOAL

For later stages we will build on top of this comparison: track-offs, comparing against modern methods, and possibly exploring some very fancy ways for feature extractions!